

CoRoT-2b

R-0195

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240717 · created 2026-07-16T21:30:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460508.8366 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.058 +/- 0.064
Transit depth (Rp/Rs)^2	0.33 +/- 0.73 [%]
Orbital Inclination (inc)	89.25 +/- 2.07
Airmass coefficient 1 (a1)	3.21 +/- 2.34
Airmass coefficient 2 (a2)	-0.97 +/- 1.1
Scatter in the residuals of the lightcurve fit is	75.23 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.086 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.058 ± 0.064 vs 0.1667 (deviation 1.7σ)
Duration fit vs published	0.086 d vs 0.095 d (deviation 1.53σ)
Mid-time O–C	-4.4 min
Depth SNR	0.9
Residual scatter	75.23 %

Light curve

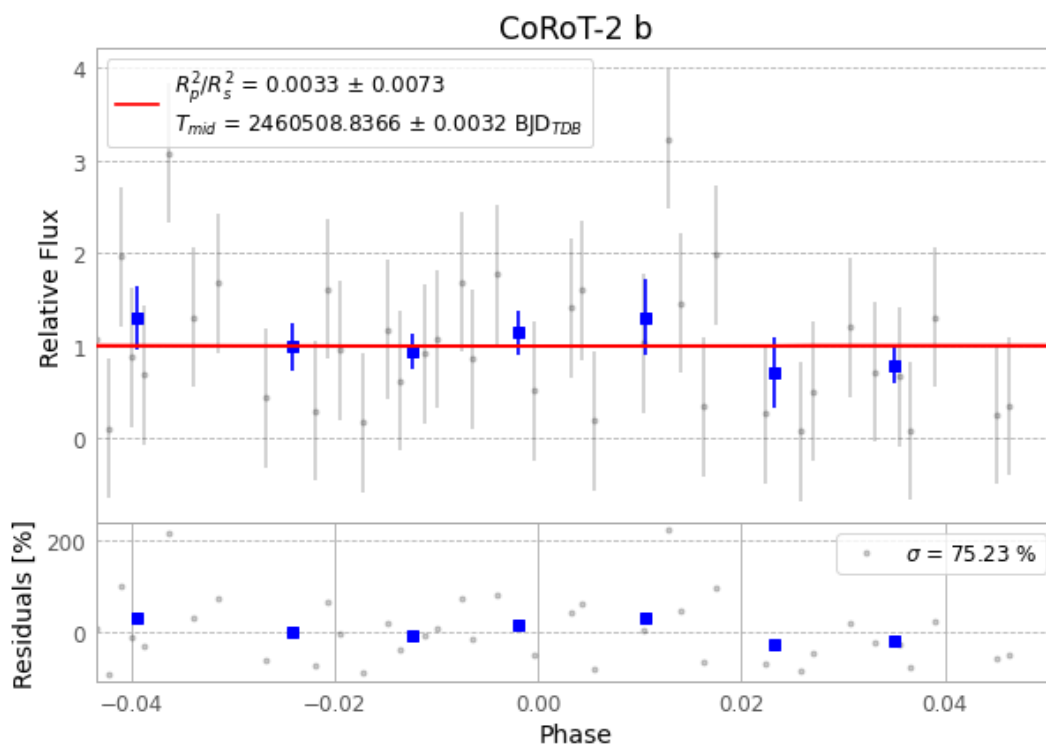


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-07-16.png (SHA-256 c16165450e1e504d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [466, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96599811d8505c6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

86 FITS frames (56 MB), CoRoT-2240717055416.FITS → CoRoT-2240717100915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240717.log (85ce6d4c467f...), FinalLightCurve_CoRoT-2 b_2024-07-16.png (c16165450e1e...), FinalLightCurve_CoRoT-2 b_2024-07-16.csv (c731ba628b04...)

Compute resources

Wall time 225.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 21:30Z.